

Juan Roberto García Gómez

MES / ERP Systems Engineer · Full-Stack Developer · Industrial IoT

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100% of injection machines, PLC stations and scanners connected	**~62%** under the approved MES budget	**~90%** less manual production reporting	**6** plant areas on one data model

PROFILE

IT Supervisor and full-stack engineer at a Korean automotive supplier (plastic injection molding and assembly). I designed, built and run the plant's in-house MES / ERP platform — from requirements to formal close, ~62% under budget — connecting 34 injection machines, 14 PLC assembly stations and 20 scanners into one data model. I lead a two-developer team and the plant's traceability program (pilot stage), and stay hands-on in React, Node.js, Python, PostgreSQL and industrial protocols (EUROMAP 63, Modbus TCP, PLC).

EXPERIENCE

IT Supervisor · Chunil México

Nov 2024 – Present

Korean automotive supplier · plastic injection molding and assembly · San Luis Potosí

MES / ERP platform — built in-house, in production

- Built the plant's MES / ERP platform in-house (React, Node.js / Express, Python, PostgreSQL, REST APIs) to replace spreadsheet control, delivering it **~62% under the approved budget**; vendors only handled legacy-PLC integration and cabling.
- Connected **100% of the plant** — **34 injection machines** (ENGEL, WOOJIN, SUMITOMO) over EUROMAP 63 and Modbus TCP, **14 assembly stations** over PLC and **20 handheld scanners** — through Python microservices (collector, packager, reporting), unifying **6 plant areas** in one data model and cutting manual production reporting by **~90%**.
- Added administrative modules now in production — HR incidences with BioTime time-clock data, production planning, warehouse and purchasing — integrated with CONTPAQi, which remains the accounting system (read-only SQL Server mirror); plus scheduled shift reports, Zebra ZPL labels and WhatsApp line-stop alerts.
- Built it for the plant floor: Socket.IO real-time dashboards, JWT access control with 20 roles, 4 Expo / React Native scanner apps, **5,300+ automated tests** (Node test runner, Vitest), and health monitoring with automatic recovery and email alerts for 20 data services.

Project management

- Ran the MES as a formal project — requirements with Production, Quality, HR and Finance, vendor coordination and budget control — and **closed it in July 2026** with a final report to plant leadership and the CEO in Korea.
- Leading the **end-to-end traceability program** (resin to finished good), now in **pilot stage**: business case and BOM for **66 planned traceability points**, vendor comparison (ScanID, Higole, Xprinter, Zebra) against external integrator quotes, and **HQ approval (Sep 2026)** of a two-stage budget — pilot, then fleet — within the reference budget. The pilot already runs on the floor with touch kiosks, label printers and in-house software.

Team and plant IT

- **Hired and now lead two developers**, managing work in Jira and code in Git / GitHub.
- Administer plant IT — Microsoft 365 tenant, Cisco Meraki MX84 firewall, managed switches, servers, NAS backups and offboarding — and delivered a **network-security assessment and upgrade proposal** to leadership.

PROJECTS

OEE Live — Real-time OEE monitoring for injection plants without an MES

KeepCoding Full Stack final project · 2026 · final grade 9.5/10

A Python collector polls 8 simulated injection machines over Modbus TCP every second; a FastAPI backend computes OEE per machine, shift and plant; a Next.js dashboard streams it live over WebSocket with downtime classification, alerts and Excel export. Deployed with Docker on Google Cloud; 298 automated tests.

Rated by the panel "the project closest to a real product in this edition."

Stack: Next.js · TypeScript · FastAPI · Python · PostgreSQL · Redis · Docker · Google Cloud

oee-live-keepcoding.cloud.run Code · GitHub

PrecioRadar — Price comparison across 21 Mexican retailers in one search

Personal product · live on Google Cloud · 2026

Queries 21 stores in parallel, matches the same product across differently named listings, recommends the best option and keeps a daily price history. Hexagonal TypeScript API (Fastify, Zod) with 111 tests, a React web app and an Expo mobile app; Playwright reads client-rendered stores. Runs on Cloud Run, Cloud SQL and Cloud Scheduler.

Stack: TypeScript · Node.js · Fastify · React · Expo / React Native · PostgreSQL · Playwright · Docker · Google Cloud

precioradar-701205043940.us-central1.run.app

PREVIOUS EXPERIENCE

Claims, Loss & Recovery Coordinator — Oct 2023 – Nov 2024

Ramírez Valle & Asociados for General Motors San Luis Potosí · San Luis Potosí

- Recovered **MX$15.2M** in logistics claims within one year by structuring case analysis and evidence management.
- Managed **100+ investigations** across claims, supplier follow-up and executive reporting.
- Built Power BI dashboards that **cut analysis time by 40%** and gave management visibility of recovery status.

EDUCATION

Full Stack Web Development Bootcamp
KeepCoding · 2025 – 2026
Completed · final project 9.5/10

Business Analytics
Tecnológico de Monterrey · 2025

Big Data, AI & Machine Learning Bootcamp
KeepCoding · 2023 – 2024

B.A. in Economics
Universidad Autónoma de San Luis Potosí (UASLP) · 2018 – 2023
Degree awarded

B.Eng. in Industrial & Logistics Engineering
Universidad Interamericana para el Desarrollo (UNID) · 2020 – 2023
Degree awarded

LANGUAGES

Spanish	Native
English	Professional working proficiency

RECOGNITION

Department Employee of the Year 2025 — Chunil México

Live projects and portfolio
Scan to open the portfolio with the live demos

robertogag.github.io

CORE SKILLS

Manufacturing systems: MES · OEE · production planning · downtime tracking · lot traceability · quality / NG tracking · ERP integration · Zebra ZPL labels

Industrial IoT: EUROMAP 63 · Modbus TCP · PLC integration (Siemens LOGO!, LS Electric) · machine data collection · real-time dashboards (Socket.IO / WebSocket)

Software: JavaScript · TypeScript · Python · React · Next.js · Node.js · Express · Fastify · FastAPI · REST APIs · React Native (Expo)

Data: PostgreSQL · SQL Server · Redis · data modeling · ETL · Power BI

DevOps & quality: Git · GitHub Actions · Docker · Google Cloud (Cloud Run, Cloud SQL) · TDD · Vitest · pytest · Playwright

IT & management: Project management · business cases · budgeting · vendor evaluation · team leadership · Jira · Microsoft 365 · Cisco Meraki · network security